Run report: run-20261014T1925Z (ecbench, Frobenius-orbit IC vs rho, fully priced)

Command. `bash research/ecbench_frobenius_orbit_vs_rho_20261014/run.sh` (eight `ecbench run --wait --cpus none --allow-busy`, then `verify --replay-all`, `table`, `compare --save`). Specs `docs/ecbench/specs/claim-koblitz-frob-orbit-full-a<a>-n<n>.json`, frozen in `PROTOCOL.md` at commit `e1708eef8`; binary sha256 `f9cfb1a09cc65074b03ee4af1aab71438ab11b2467391abcfed1594a98484d19` built from that commit. Start 2026-10-14T19:25:54Z (UTC).

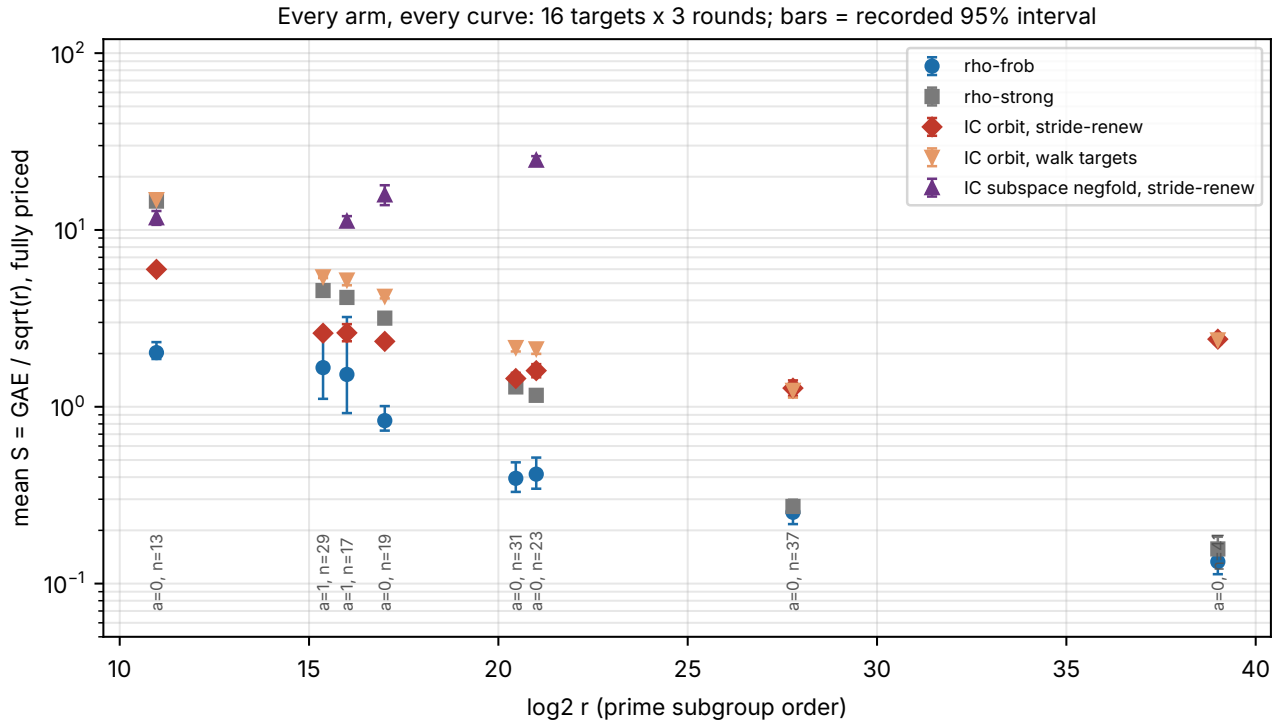
Host and isolation. Intel Xeon @ 2.80 GHz, 4 vCPU, shared container; L0 on every record (no CPU reservation). Wall times are descriptive only.

Workloads. Eight Koblitz curves, 16 public targets each (target seeds 202610142001–202610142015, odd), 3 measured rounds + 1 warm-up, alternate order. Arms: rho-frob (`rho.signed_frobenius_full`, reference), rho-frob-aa (A/A control), rho-strong (`rho.signed_frobenius_strong_full`), `ic-orbit-c<c>-renew` and `-walk` (`ic.pipeline_full`, compact-orbit-scan, mitm-frobenius-counted:m=2), `ic-negfold-d<d>-renew` (n ≤ 23 only).

Completion. 2,816 records requested, 2,816 written, 2,816 verified; no failures, no timeouts. `verify --replay-all`: 2,112 of 2,112 measured records replayed identically; every audit `ok`. Two sessions (a1-n17, a1-n29) record `git_dirty = true` (whitespace-only rustfmt of source during the run; binary unchanged).

Measured values (mean S = GAE/√r over 48 measured runs per arm, 95 % interval as `ecbench table` recorded; every record `lower_bound = false`). Unit: `ecbench.gae`.

session	curve	rho-frob	rho-strong	orbit renew	orbit walk	negfold renew
ECBS1hb67a421a6fe2	a0-n13	2.025 [1.863, 2.321]	14.562 [14.255, 15.026]	5.986 [5.857, 6.116]	14.847 [14.672, 15.052]	11.718 [10.688, 12.801]
ECBS1h2627f1e2aee2	a1-n17	1.524 [0.921, 3.218]	4.153 [4.026, 4.310]	2.619 [2.348, 2.932]	5.210 [4.867, 5.586]	11.214 [10.501, 11.972]
ECBS1h8baa27b989e6	a0-n19	0.835 [0.733, 1.009]	3.172 [3.084, 3.262]	2.343 [2.233, 2.478]	4.217 [4.100, 4.362]	15.786 [13.819, 17.902]
ECBS1h5ace8f3e67d1	a0-n23	0.416 [0.344, 0.516]	1.161 [1.109, 1.215]	1.601 [1.468, 1.742]	2.126 [1.992, 2.266]	24.807 [23.361, 26.152]
ECBS1h575c8bcdc35a	a1-n29	1.665 [1.109, 2.639]	4.543 [4.454, 4.644]	2.607 [2.532, 2.699]	5.431 [5.346, 5.528]	–
ECBS1ha274653650f8	a0-n31	0.394 [0.330, 0.485]	1.295 [1.245, 1.350]	1.445 [1.342, 1.555]	2.163 [2.055, 2.286]	–
ECBS1h7aed5aa2d512	a0-n37	0.253 [0.217, 0.290]	0.273 [0.248, 0.298]	1.275 [1.153, 1.410]	1.239 [1.127, 1.350]	–
ECBS1h868a2da9acdc	a0-n41	0.133 [0.113, 0.154]	0.157 [0.132, 0.184]	2.412 [2.301, 2.526]	2.390 [2.267, 2.523]	–



Sessions: research/ecbench_frobenius_orbit_vs_rho_20261014/sessions/*; ecbench table

Figure 1: Mean S per arm and curve with the recorded 95 % intervals; sessions as in the table.

Controls. rho-frob-aa ÷ rho-frob = 1.000 [1.000, 1.000] on every curve (`tables/compare-*-rho-frob-aa-vs-rho-frob.txt`).

Raw records. `sessions/<curve>/records.jsonl`; audits `audits/audit-<curve>.json`; tables and comparisons `tables/`, `sessions/<curve>/comparisons/`. Interpretation is in the study README and REPORT.pdf.